Warm-up / Why did fruit sales rise?

Separate a change from its cause

Name: _____ Date: _____

Suggested lesson time: 30 minutes. Work individually, then compare when invited.

Useful words

Cause: Something that helps produce an outcome.

Comparison group: A group used to judge what might happen without the change being tested.

Random assignment: Using chance to decide which groups receive a change, helping avoid systematic differences.

Read the scenario

Invented scenario: A cafeteria sells more fruit in the week it displays healthy-eating posters. That week it also cuts fruit prices. The manager says the posters caused the rise.

Produce: Write an observation, another possible cause, and one sentence explaining why the posters have not been isolated.

Your steps

1. State what happened without naming its cause.
2. Name the other change and explain how it could affect sales.
3. Explain why these facts cannot establish the effect of posters alone.

Before you finish: Have you kept an observed rise separate from an explanation for it?

Your response

Main activity / Lighting, patrols, and theft reports

Name: _____

Read the scenario

Invented scenario: Reported thefts on a street fall from 50 in September to 35 in October. Brighter lights are installed at the start of October, and extra evening patrols begin that same day. The mayor says the lights caused the drop.

Produce: Write the observation, a competing explanation, and a proposed comparison in three labelled parts.

Your steps

1. State the observed change separately from the mayor's causal explanation.
2. Name the other change that could explain some or all of the drop.
3. Propose a comparison that would help separate the effects: describe which streets to compare and which other condition should be kept similar. Do not claim your proposed comparison has already been done.

Before you finish: Does the comparison vary lighting without also giving only the newly lit streets extra patrols?

Your response

Exit ticket / Why did absences fall?

Name: _____

Read the scenario

Invented scenario: Weekly absences in a school fall from 12 to 5 after text reminders and a new bus service begin on the same day. The principal credits the reminders alone.

Produce: Write the observed change, another possible cause, and a comparison that could test reminders.

Your steps

1. State the observed change without attributing a cause.
2. Name the other change and explain its possible effect.
3. Describe groups to compare and which other condition must stay comparable.

Before you finish: Would your comparison give similar bus access to students with and without reminders?

Your response